aNLP | Wordsmiths

DiffusEdit

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Abstract

Reference articles go out of date as events develop, and updating them by hand is slow. Using a language model to automate the update rewrites text that was already correct and leaves no record of what changed. DiffusEdit instead edits the article in place. The frozen model rewrites only the stale positions, and the system repeats the process on sentences that the rewrites themselves made wrong.

Problem

- **Reference text ages rapidly:** Real-world events (news, Wikipedia) constantly makes existing facts out-dated.
- **Current LLM solutions are brute-force:** Auto-regressive generation rewrites the entire article from scratch, which is slow and blindly overwrites perfectly correct sentences.
- **The "Cascading Edit" Dilemma:** Facts are logically connected. If you change a fact in paragraph one, sentences in paragraph three might suddenly become completely unsupported.

Example:

Sentence 1: “Three people died”

Sentence 2: “The small death toll limited the political fallout“

A week later, let’s say the death toll increases to 42.

S1: wrong
S2 : wrong by dependence

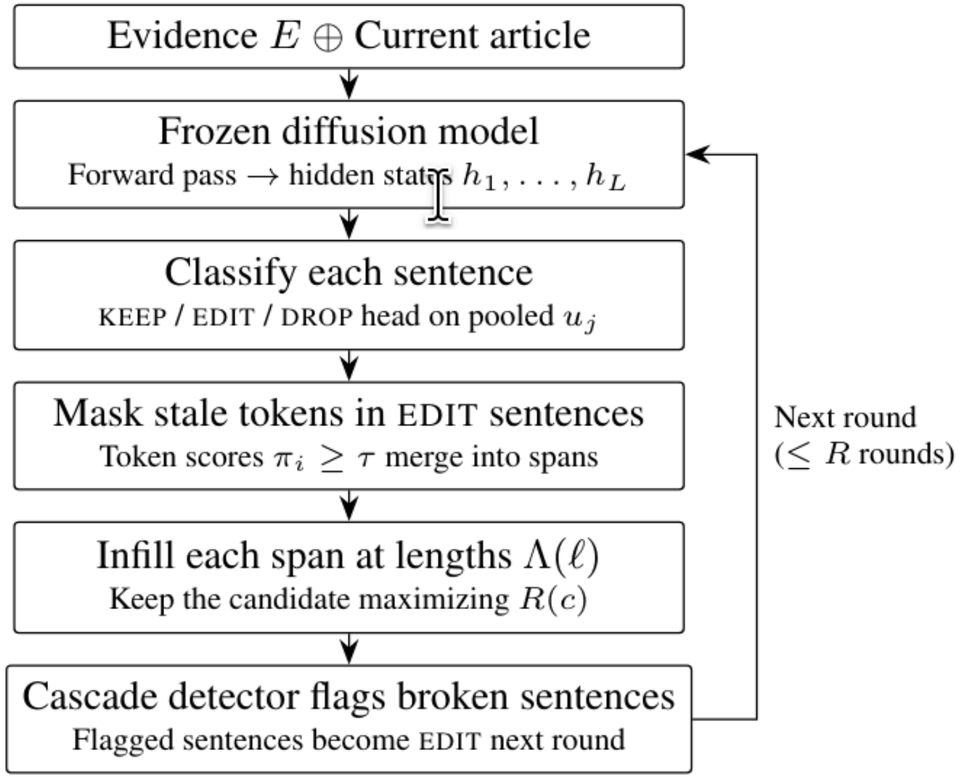
Contributions

- A **sentence-level KEEP/EDIT/DROP classifier** over pooled hidden states, which restricts token scoring to EDIT sentences and adds deletion through DROP.
- An iterative edit loop that returns broken sentences to the pipeline, comparing three **cascade detectors**: mention overlap, staleness delta, and a learned consequence head.
- **Variable-length repair through multi-length infilling** and reranking for fluency and evidence support.
- **Sentence- and token-level staleness labels** derived from the dataset by alignment, plus a mined and hand-checked set of cascade pairs.
- An **evaluation** separating detection quality, update quality, minimality, and cascade recall, with baselines, oracles, and ablations.

Dataset

- To train our classifiers, we used the FRUIT dataset, which provides paired Wikipedia snapshots (outdated vs. updated) alongside the new evidence.
- Because FRUIT did not have manual, sentence-level annotations, we used text-alignment heuristics to automatically label sentences as KEEP, EDIT, or DROP.
- However, this method led to a HUGE problem: Wikipedia editors not only update facts but also perform concurrent stylistic rewrites such as switching from active to passive voice. Our algorithmic heuristic blindly flags these as factual EDITs.
- Our frozen diffusion backbone (LLaDA-8B) encodes text based on the semantics of the facts rather than just surface-level grammar and syntax.
- A stylistic rewrite has the exact same hidden-state vector as a factually clean KEEP sentence.
- This semantic collision causes ?? label noise rate across the dataset. This diminishes the theoretical upper bound of our linear classifiers.

Pipeline


Evidence $E \oplus$ Current article
Frozen diffusion model
Forward pass $\rightarrow$ hidden states $h_1, \ldots, h_L$
Classify each sentence
KEEP / EDIT / DROP head on pooled u_j
Mask stale tokens in EDIT sentences
Token scores $\pi_i \geq \tau$ merge into spans
Infill each span at lengths $\Lambda(\ell)$
Keep the candidate maximizing $R(c)$
Cascade detector flags broken sentences
Flagged sentences become EDIT next round
Next round
($\leq R$ rounds)

Current Progress

Feature Extraction

Masking and Variable Length Infilling

Token Staleness Identification

Sentence-Level Classifier

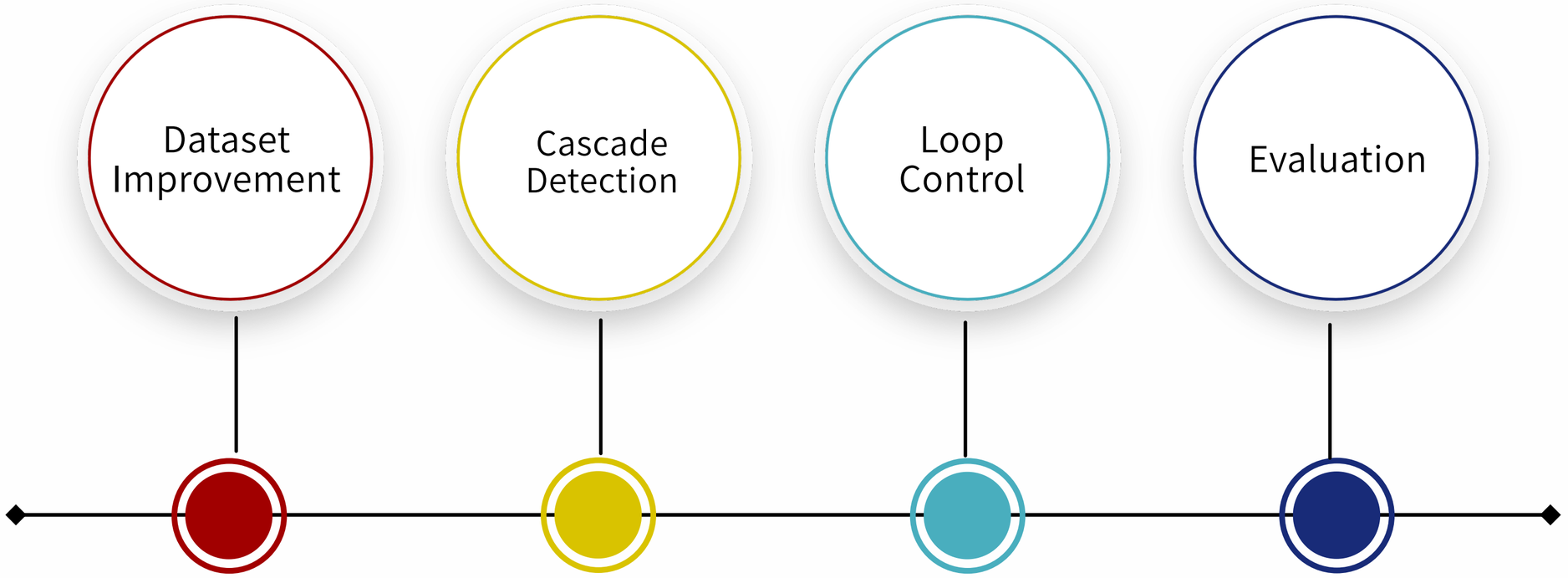
Ablations:

- Non-diffusion baselines
- Sentence-Level Classifier with evidence

Results

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Next Steps


Dataset Improvement
Cascade Detection
Loop Control
Evaluation

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Questions?

Your doubt is [MASK] - we'll infill it

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THANK YOU

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